

End Sem Exam

● Graded

Student

PRAGATI AGRAWAL

Total Points

99 / 100 pts

Question 1

True or False

9 / 10 pts

– 0 pts Correct

✓ – 1 pt Click here to replace this description.

– 2 pts Click here to replace this description.

– 3 pts Click here to replace this description.

– 5 pts Click here to replace this description.

– 4 pts Click here to replace this description.

Question 2

TLS, Tor, PGP, MCQs

20 / 20 pts

✓ + 20 pts Correct

– 2 pts Incorrect (A)

– 2 pts Incorrect(B)

– 2 pts Incorrect(c)

– 2 pts Incorrect(D)

– 2 pts Incorrect (E)

– 2 pts Incorrect(F)

– 2 pts Incorrect(G)

– 2 pts Incorrect(h)

– 2 pts Incorrect (I)

– 2 pts Incorrect(J)

Question 3

Parity Bits

9 / 9 pts

– 5 pts Correct

– 2 pts Click here to replace this description.

– 2 pts Click here to replace this description.

✓ – 0 pts Click here to replace this description.

Question 4

Longest Prefix

17 / 17 pts

– 3 pts Correct

– 3 pts Click here to replace this description.

– 3 pts Click here to replace this description.

– 4 pts Click here to replace this description.

– 4 pts Click here to replace this description.

✓ – 0 pts Click here to replace this description.

Question 5

Dijstras

10 / 10 pts

– 5 pts Incorrect answer for (a)

– 5 pts Incorrect answer for (b)

– 0 pts Incorrect explanation for (a) / No explanation provided

– 0 pts Incorrect explanation for (b) / No explanation provided

✓ + 10 pts Click here to replace this description.

Question 6

MAC

10 / 10 pts

✓ – 0 pts Correct

– 2 pts Click here to replace this description.

– 8 pts Click here to replace this description.

– 4 pts Click here to replace this description.

Question 7

Learning Switches

12 / 12 pts

✓ – 0 pts Correct

– 4 pts Click here to replace this description.

– 2 pts Click here to replace this description.

– 6 pts Click here to replace this description.

– 0 pts Click here to replace this description.

– 7 pts Click here to replace this description.

– 3 pts Click here to replace this description.

Question 8

8a) Crypto

8 / 8 pts

✓ - 0 pts Correct

- 2 pts One Wrong Option

- 4 pts Partially Correct

- 6 pts One Correct Option

- 8 pts Complete Wrong

Question 9

8b) Crypto 2

4 / 4 pts

✓ - 0 pts All

- 4 pts Nothing



Indian Institute of Technology, Kanpur
Computer Science and Engineering

End Semester Exam
CS425: Computer Networks

Instructor: Adithya Vadapalli

04/05/2025

Name: PRAGATI AGRAWAL

Roll Number: 220779

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1. This exam contains 12 pages (including this cover page) and 8 questions.
 2. Total of marks is 100.
 3. You are allowed to bring one A4-sized cheat sheet, which is handwritten (by you!!) notes to the exam.
 4. Please write your name in **ALL CAPS**. You have **120 minutes** to solve the exam.
 5. Good luck!

Distribution of Marks

Question	Points	Score
1	10	
2	20	
3	9	
4	17	
5	10	
6	10	
7	12	
8	12	
Total:	100	

1. This question has 10 parts. Each of them is a true or false question. No need to give an explanation. There are no negative marks for wrong answers.

(a) (1 point) NAT allows multiple devices in a private network to share a single public IP address when accessing the Internet.

True

(b) (1 point) NAT devices change both source and destination IP addresses in a packet.

~~True~~ False

(c) (1 point) The Cyclic Redundancy Check (CRC) can detect all single-bit errors.

False

(d) (1 point) ALOHA protocol has a higher maximum efficiency than Slotted ALOHA.

False

(e) (1 point) TLS operates between the transport and application layers and provides end-to-end encryption.

True

(f) (1 point) WEP provides strong authentication for Wi-Fi networks.

False

(g) (1 point) Diffie-Hellman is vulnerable to man-in-the-middle attacks if not authenticated.

True

(h) (1 point) In TLS, the client authenticates the server using a digital certificate.

True

(i) (1 point) Network Layer ensures that packets are transported reliably over an unreliable channel.

~~True~~ False

(j) (1 point) Link State routing protocols use the Bellman-Ford algorithm to compute shortest paths.

False

2. Transport Layer Security (TLS), Tor, PGP, and SSH are widely used tools and protocols for securing communication. Answer the following questions. Color the box adjacent to the correct answer(s). For some questions, more than one answer could be correct. You have to mark all the correct answers to get credit.

(a) (2 points) What are the primary security goals of TLS? (Choose all that apply)

- ☒ Authentication
- ☐ Non-repudiation
- ☒ Confidentiality
- ☒ Integrity
- ☐ Availability

Ans = Authentication, Confidentiality, Integrity

(b) (2 points) In the TLS handshake, what type of key is typically used to encrypt the communication **after the handshake is complete**?

- ☐ Public key
- ☒ Private key
- ☒ Symmetric key
- ☐ None of the above

Ans = Symmetric Key

(c) (2 points) How does TLS ensure the **integrity** of messages during communication?

- ☒ By using hashing algorithms and message authentication codes (MACs)
- ☐ By using public key encryption
- ☐ By using digital certificates
- ☐ By performing the handshake

Ans = ↑

(d) (2 points) What role do **digital certificates** play in the TLS handshake?

- ☐ They encrypt the data
- ☒ They verify the server's identity
- ☐ They provide a shared secret key
- ☐ They prevent the use of weak encryption algorithms

Ans

(e) (2 points) What is the primary purpose of the **Tor** network?

- ☐ To accelerate internet speed using multiple paths
- ☐ To block malware and phishing attacks

- ☐ To perform encrypted email communication
- ☒ To enable anonymous communication over the internet *Ans*

(f) (2 points) In the Tor network, what is the correct sequence of nodes that traffic passes through?

- ☐ Middle relay → Entry node → Exit node
- ☐ Exit node → Entry node → Middle relay
- ☒ Entry node → Middle relay → Exit node *Ans*
- ☐ Middle relay → Exit node → Entry node

(g) (2 points) What is the main function of PGP in secure communications?

- ☐ Preventing denial-of-service attacks
- ☒ Encrypting and signing email messages *Ans*
- ☐ Establishing a VPN connection
- ☐ Preventing data exfiltration

(h) (2 points) Which of the following is a core security property provided by Off-the-Record (OTR) Messaging?

- ☐ Message authenticity using digital signatures that can be verified later
- ☒ Perfect forward secrecy and deniable authentication *Ans*
- ☐ Centralized key management for user identities
- ☐ Long-term non-repudiation for audit purposes

(i) (2 points) Which of the following best describes the primary use of SSH?

- ☐ Accessing web pages securely
- ☒ Secure remote login to a server *Ans*
- ☐ Encrypting file systems
- ☐ Creating anonymous communication tunnels

(j) (2 points) During the initial SSH handshake, how is the server's identity typically established?

- ☐ Symmetric key exchange
- ☒ Challenge-response over plaintext *X*
- ☒ Server's public key authentication *ans = Server's public key authentication*
- ☐ DNS over HTTPS

3. A 4×4 block of data was transmitted using even parity in both row and column directions. The original data and its corresponding parity bits were arranged in the following 5×5 matrix. The last row and column contain the parity bits for the columns and rows, respectively; the bottom-right cell is the overall parity bit:

(Note: The vertical bar separates the row data from the row parity bit. The last row contains column parity bits and the overall parity bit. The horizontal bar separates the column data from the column parity bit.)

Upon reception, the following matrix was observed:

	Column 1	Column 2	Column 3	Column 4	Row Parity
Row 1	1	0	1	1	1
Row 2	0	1	1	0	1 ← 0
Row 3	1	1	1	0	1
Row 4	0	0	1	1	0
Column Parity	0	1 ↑	0	0	1

- (a) (5 points) Identify the position (which row and column) of the error using 2-D parity analysis. Correct the error and write the corrected 4×4 data block.

The error is row 2 col 2. Because row parity is written 1 but it is coming 0. And col parity also the same problem. The correct bit should be 0 on row 2, col 2.
Corrected block data:

```

1 0 1 1
0 0 1 0
1 1 1 0
0 0 1 1

```

- (b) (2 points) Can single-bit errors be corrected using single-dimensional parities?

No, they can be detected but not corrected

- (c) (2 points) Can two-bit errors be corrected using two-dimensional parities?

No, 2D errors can be detected but not corrected using 2D parity.

4. Consider a datagram network using **12-bit host addresses**. A router uses **longest-prefix matching** to determine the appropriate interface for forwarding. The router has the following forwarding table:

Prefix Match (binary)	Interface
1	1
110	2
1101	3
11011	4
010	5
0100	6
0001	7
otherwise	8

Answer the following questions based on this table:

- (a) (3 points) A datagram arrives at the router with destination address 110110101011. To which interface will this datagram be forwarded?

4

- (b) (3 points) A datagram arrives at the router with destination address 010011001100. To which interface will this datagram be forwarded?

6

- (c) (3 points) A datagram arrives at the router with destination address 111000110101. To which interface will this datagram be forwarded?

1

- (d) (4 points) Provide an example of a 12-bit destination address that would match both the prefix 1101 and 110, and explain (in one short sentence) why the router chooses the interface it does.

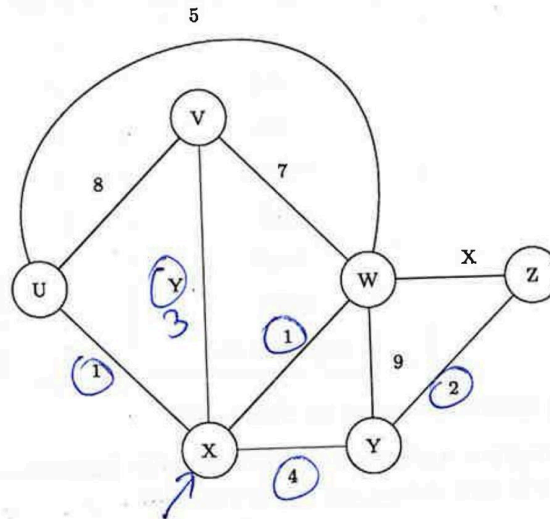
110101010101. Using Longest Prefix Match, length 4 match will be given priority over length 3 match. So it will be forwarded to interface 3. ~~If 5th bit does not match~~

- (e) (4 points) Suppose the prefix 0100 is removed from the table. How would that affect the forwarding decision for a datagram with destination address 010011111000?

It will now be forwarded to interface 5 earlier it was forwarded to interface 6.

Since 5th bit does not match for row 4 in the table, this address can't be forwarded to interface 4 either! So it will be forwarded to interface 3.

5. Consider the incomplete 6-node network shown below, with the given link costs.



Completed table for shortest distance to all nodes from X using Dijkstra's algorithm:

Node	Shortest distance from X	Previous Node
X	0	n/a
U	1	X
W	1	X
V	3	X
Y	4	X
Z	6	Y

- (a) (5 points) For link X, what is the cost associated with this link? If the answer can't be determined given the information, respond with n/a.

n/a. Since link X is never used, we can't determine its cost. We can say that it must be greater than 4 otherwise $X \rightarrow W \rightarrow Z$ would have been a shorter route to Z but since $X \rightarrow Y \rightarrow Z$ is the shortest route, $X > 4$.

- (b) (5 points) For link Y, what is the cost associated with this link? If the answer can't be determined given the information, respond with n/a.

Y's cost must be 3 because it is used to go from X to V and previous node is X itself, which means it is the ~~short~~ direct link connecting these two. $Y = 3$

6. For this question, look at Figure 1. The IP and MAC addresses are shown for nodes A, B, C, and D, as well as for the router's interfaces. Consider an IP datagram being sent from node D to node C.

- (a) (2 points) What is the source MAC address at point 5?

EE-DC-84-BE-11-21

- (b) (2 points) What is the destination MAC address at point 5?

EF-F1-9F-29-EA-7B

- (c) (2 points) What is the source IP address at point 5?

128.119.37.136

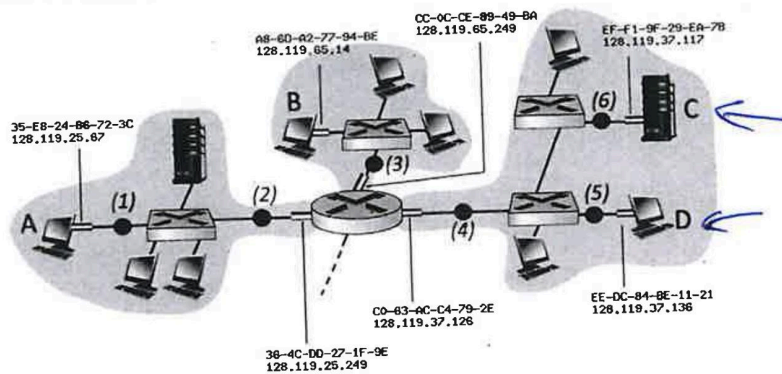


Figure 1: The Network Setup

- (d) (2 points) What is the destination IP address at point 5?

128.119.37.117

- (e) (2 points) Do the source and destination MAC addresses change at point 6? Answer with yes or no.

No

7. A learning switch has four ports and starts with an empty forwarding table. It receives the following sequence of Ethernet frames:

Frame #	Source MAC	Destination MAC	Port Received On
1	A1	B1	1
2	B1	A1	2
3	C1	B1	3
4	D1	E1	4

- (a) (4 points) After processing all four frames, what is the final state of the switch's forwarding table? Fill in the table below: (1 mark for each row).

MAC Address	Port
A1	1
B1	2
C1	3
D1	4

- (b) (8 points) For each frame, state whether it will be **flooded** or **forwarded**, and if forwarded, to which port. (2 marks for each row. A row must be completely correct to receive credit. For example, if the action "Forwarded" is correct but the port to which it is forwarded is incorrect, you will receive 0 out of 2 marks for that row.)

Frame #	Action (Flooded / Forwarded to Port X)
1	Flooded
2	Forwarded to Port 1
3	Forwarded to Port 2
4	Flooded

8. Alice wants to send a secure message M to Bob over an insecure channel. She wants to ensure the following:

- Only Bob can read the message (**Confidentiality**)
- Bob can verify that Alice sent the message (**Authentication**)
- Bob can detect if the message was tampered with (**Integrity**)

Assume:

- Alice has Bob's public key: K_B^{pub}
- Bob has Alice's public key: K_A^{pub}

- (a) (8 points) Match each of Alice's steps with its correct description.

Step	Description	Purpose (Match)
A	Compute $h = \text{Hash}(M)$	(i) Provide confidentiality
B	Compute $\sigma = \text{Sign}_{K_A^{\text{priv}}}(h)$	(ii) Provide authentication
C	Compute $C = \text{Enc}_{K_B^{\text{pub}}}(M, \sigma)$	(iii) Send the encrypted message over network
D	Send ciphertext C to Bob	(iv) Check integrity

Answer: A-(iv), B-(ii), C-(i), D-(iii)

- (b) (4 points) Bob receives a ciphertext C that contains a message M and a digital signature σ . The table below lists four steps Bob may perform to process the ciphertext.

Step	Description
A	Compute $h' = \text{Hash}(M)$
B	Extract M and σ
C	Verify $\text{Verify}_{K_A^{\text{pub}}}(\sigma, h')$
D	Decrypt C using K_B^{priv}

What is the correct order in which Bob should perform the above steps to verify the authenticity and integrity of the message? Write the correct sequence of step labels (e.g., $A \rightarrow B \rightarrow \dots$). You will not receive any partial credit in this question. It is all or nothing.

$D \rightarrow B \rightarrow A \rightarrow C$

